4.a) - Data Modelling and Form Design

Aim

To design and develop a **Blood Donor & Camp Management System** using Zoho Creator by creating structured forms, applying appropriate validations, generating reports, and managing donor registrations, blood donation camps, donations, blood inventory, and blood requests.

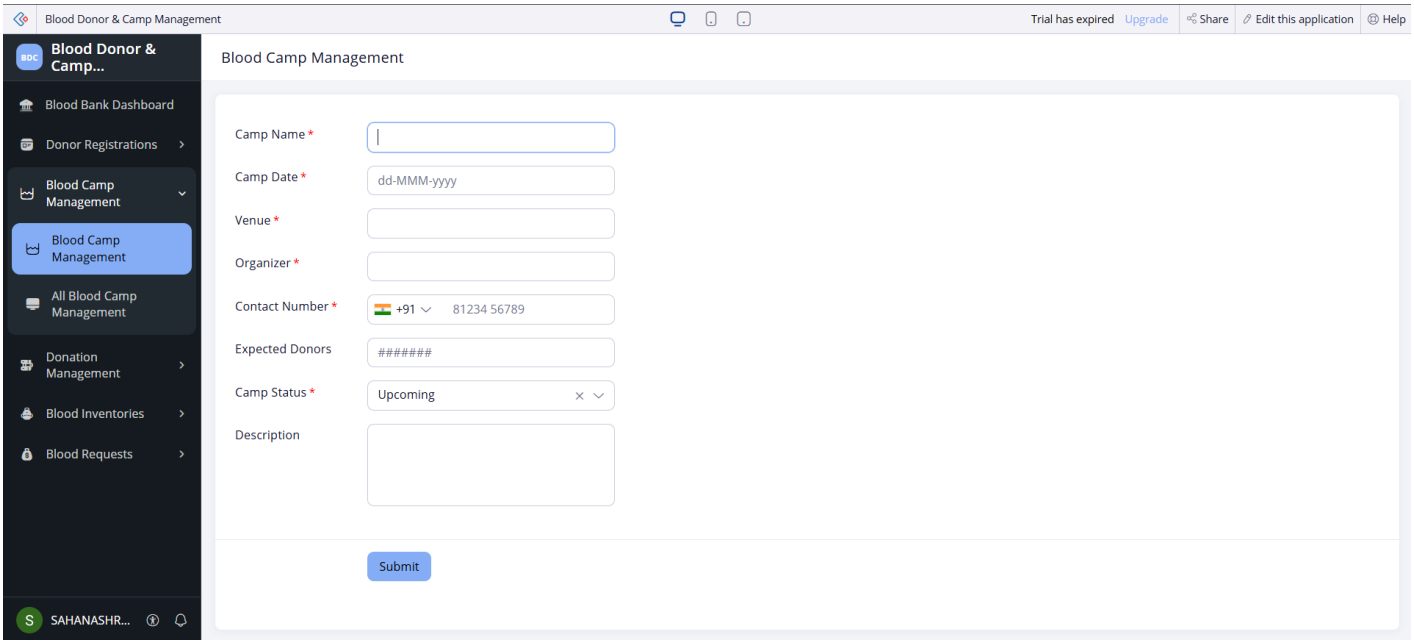
Theory

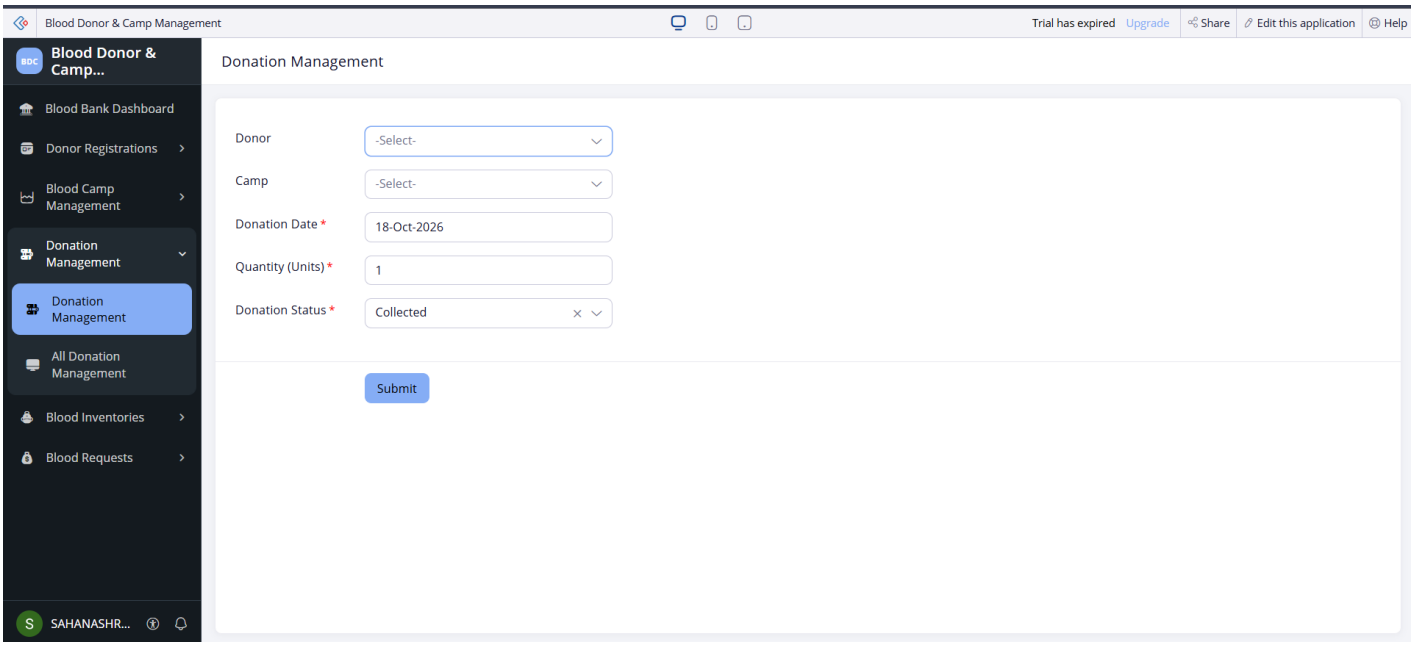
1. Data modelling and form design involve organizing data into suitable fields and forms for efficient collection, storage, and management. In Zoho Creator, forms can be created using different field types such as Single Line, Multi Line, Number, Date, Phone, Email, Dropdown, and Lookup.
2. In this experiment, a **Blood Donor & Camp Management** application is developed to manage blood donation activities through five forms: Donor Registration, Blood Camp Management, Donation Management, Blood Inventory, and Blood Requests.
3. The **Donor Registration** form collects donor details, including name, date of birth, gender, blood group, contact information, address, eligibility status, and donor status. The **Blood Camp Management** form stores camp details such as camp name, date, venue, organizer, contact number, expected donors, and camp status. The **Donation Management** form records donations by linking donors and camps with donation dates, quantities, and donation status.
4. The **Blood Inventory** form maintains the available units for different blood groups and records the last updated date. The **Blood Requests** form stores hospital and patient request details, required blood group, units required, urgency, request status, and units issued.
5. Validations are applied to improve data accuracy, including mandatory fields, appropriate field types, valid quantity values, and checks to ensure that units required are greater than zero, units issued are not negative, and units issued do not exceed units required. The submitted records are displayed in their respective reports for easy viewing and management.
6. A dashboard is also created to provide an overview of donor information through a total donor count, charts, and a Blood Requests report. The application supports organized record management and demonstrates how a low-code platform can be used to develop a centralized blood donation management system.

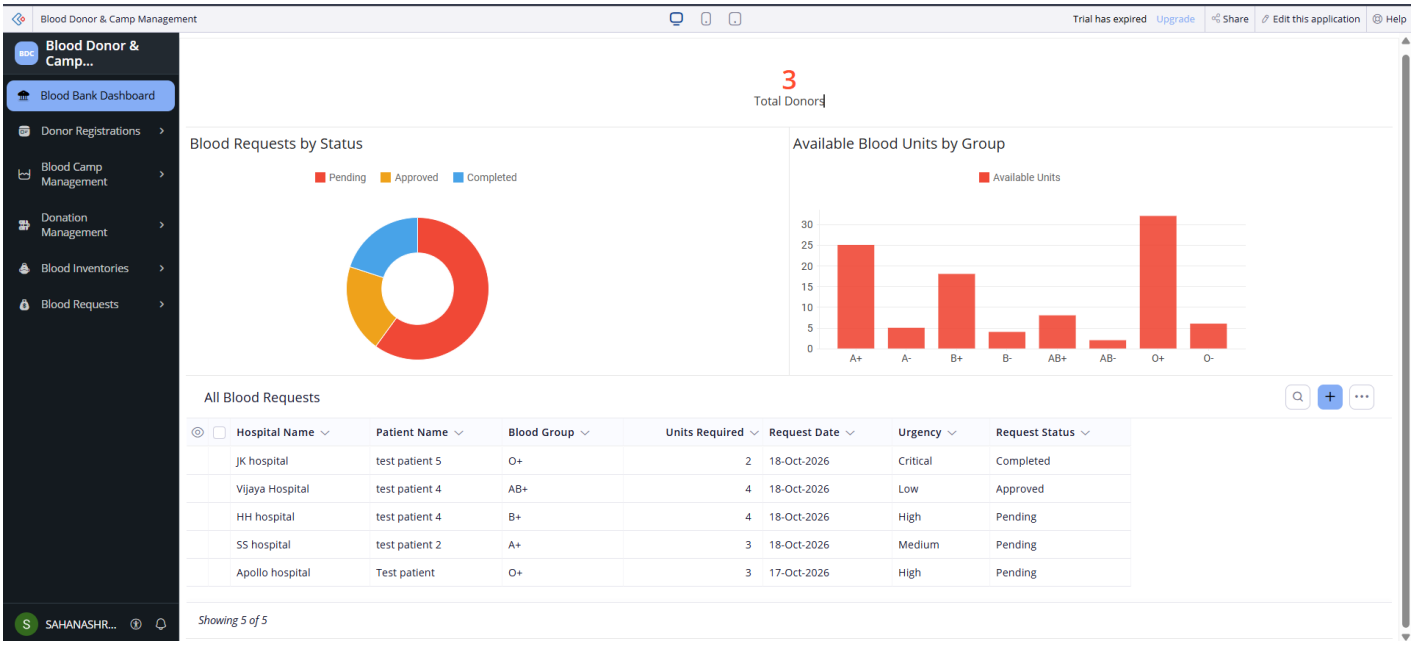
Procedure

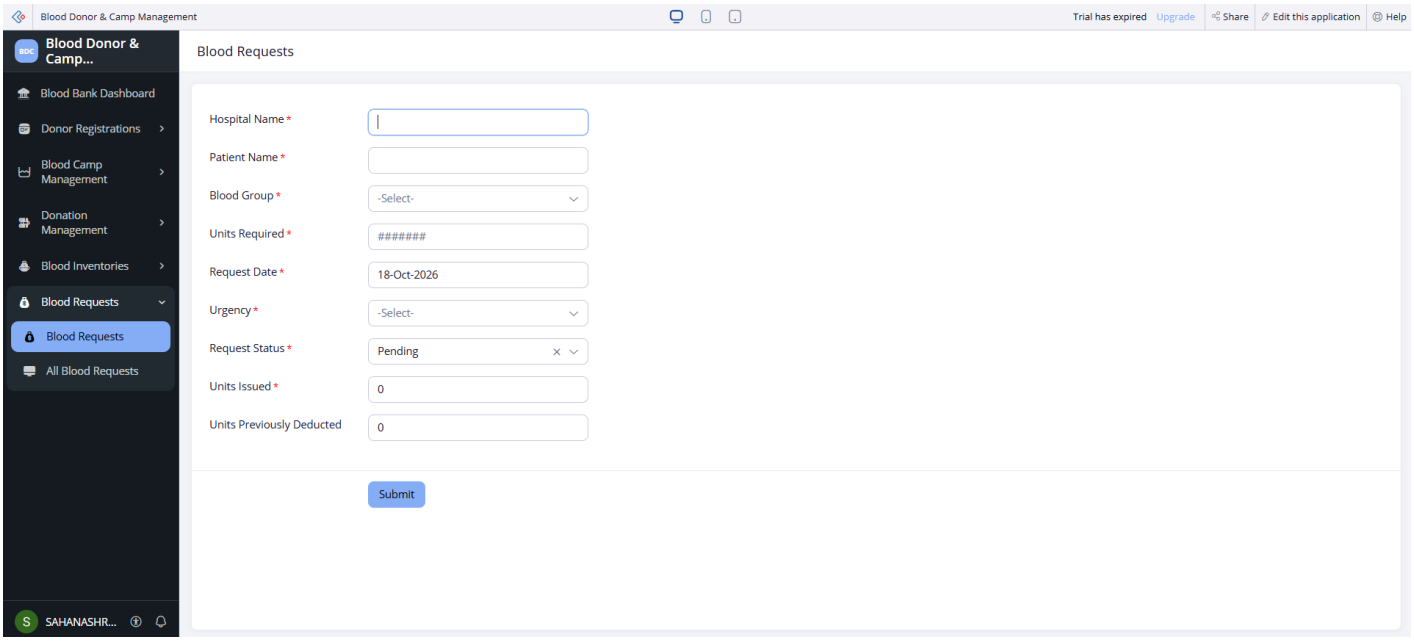
1. Open Zoho Creator and create the **Blood Donor & Camp Management** application.
2. Create the **Donor Registration** form and add the required donor information fields.
3. Create the **Blood Camp Management** form and add camp details, including date, venue, organizer, and status.
4. Create the **Donation Management** form and link donation records to the corresponding donor and blood camp using Lookup fields.

5. Create the **Blood Inventory** form to maintain blood groups, available units, last updated date, and inventory notes.
6. Create the **Blood Requests** form to record hospital requests, required blood groups, units required, urgency, request status, and units issued.
7. Configure the appropriate field types, mandatory fields, dropdown options, and validation rules.
8. Create reports for all five forms to display and manage the submitted records.
9. Create a dashboard using Page Builder with a total donor count panel, charts, and the Blood Requests report.
10. Enter sample records, preview the application, and test the forms and validation rules.
11. Verify that the submitted records appear correctly in their respective reports and that the dashboard displays the available information.









Result

The **Blood Donor & Camp Management System** was successfully developed in Zoho Creator with five structured forms, appropriate field validations, reports, and a dashboard. Sample records were submitted and displayed in their respective reports. The Blood Requests validation workflow was tested successfully, demonstrating the prevention of invalid quantity entries. The application provides a centralized interface for managing donor registrations, blood donation camps, donation records, inventory information, and blood requests.